

The 65th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

722.ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION

Lymphocyte Reconstitution after Naïve T Cell-Depleted Hematopoietic Cell Transplantation

Melinda Ann Biernacki, MD¹, Kyle B Woodward, PhD¹, Hugh R. MacMillan¹, Madhavi Lakkaraja, MDMPH^{2,3}, Diego Archila-Diaz, PhD¹, Reema Jain, PhD¹, Heather Persinger¹, Barbara Hilzinger, RN¹, Jacqueline Diaz¹, Warren Shlomchik, MD⁴, Evan W. Newell, PhD¹, Marie Bleakley, MD PhD^{1,5}

¹ Fred Hutchinson Cancer Center, Seattle, WA

² Department of Pediatrics, Division of Hematology/Oncology, University of Washington School of Medicine, Seattle, WA

³ Clinical Research Division, Fred Hutchinson Cancer Center, Seattle, WA

⁴ Startzl Transplant Institute, Pittsburgh, PA

⁵ Division of Pediatric Hematology, Oncology, Bone Marrow Transplant & Cellular Therapy, Seattle Children's Hospital, Seattle, WA

Background: Allogeneic hematopoietic cell transplantation (HCT) is curative for many high-risk hematological malignancies. However, graft-versus-host disease (GVHD) causes morbidity, mortality, and reduced quality of life after HCT. Removal of all T cells from hematopoietic cell grafts (pan-T cell depletion; pan-TCD) significantly reduces GVHD but is associated with impaired immune reconstitution and increased non-relapse mortality due partly to opportunistic infections. We developed a novel strategy to remove CD45RA⁺ naïve T cells (T_N) from peripheral blood stem cell grafts (PBSC) that is associated with a low incidence of serious acute GVHD (aGVHD) and an exceptionally low incidence of chronic GVHD (cGVHD) (Bleakley, JCO 2022). Although CD45RA-depletion removes T_N and other CD45RA-expressing effector T cell populations from PBSC along with most NK and B cells, T_N-depletion (T_ND) does not appear to increase infections or have a major negative impact on immune reconstitution. Understanding subtle differences in lymphocyte population dynamics after T_ND compared to T cell-replete (TR) HCT will enable further graft engineering advances and may provide broader insights into human HCT immunobiology.

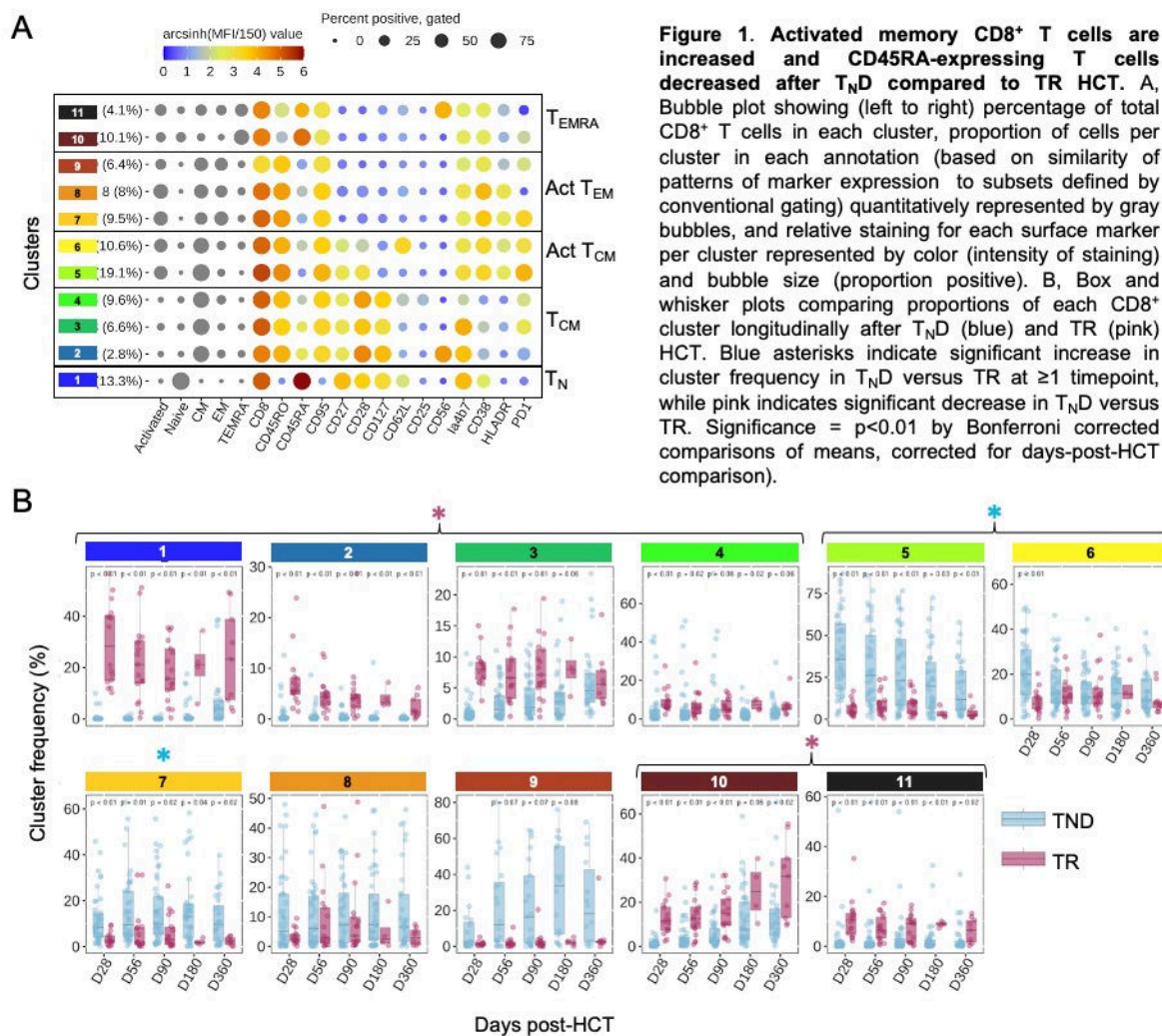
Objective: To evaluate lymphocyte reconstitution after T_ND HCT.

Methods and Results: We compared blood lymphocyte populations in T_ND HCT recipients (n=47) to those in TR HCT controls (n=19) at days 28, 56, 90, 180, and 360 post-HCT using multiparametric flow cytometry. To complement standard Boolean analysis, we employed a metaclustering approach to simultaneously evaluate ~20 cell surface markers for each class of lymphocytes (CD8⁺ and CD4⁺ T cells, NK cells, and B cells) over scores of batches of recipient and control samples. Absolute numbers of total CD8⁺ T cells were similar in T_ND compared to TR at all timepoints, whereas total CD4⁺ T cell numbers were consistently lower in T_ND until 1 year post-HCT. Using conventional gating and high-dimensional clustering analyses, we identified several key differences in T cells after T_ND. CD45RA-expressing populations (T_N and effector memory re-expressing CD45RA [T_{EMRA}]) in CD8⁺ and CD4⁺ T cells were rare in CD45RA-depleted PBSC and remained decreased in T_ND HCT recipients until around 1 year (CD8⁺, Fig. 1). Activated CD8⁺ and CD4⁺ central and effector memory (T_{CM}, T_{EM}) T cells were increased proportionally and in absolute numbers in T_ND early post-HCT (CD8⁺, Fig. 1). Absolute numbers of regulatory T cells (Treg) were decreased in T_ND; consequently, ratios of activated CD8⁺ and CD4⁺ to Treg were both markedly increased early post-HCT, despite a similar incidence of aGVHD in T_ND and TR recipients. Although most NK cell populations were removed from CD45RA-depleted PBSC, absolute numbers of NK cells in all compartments, including both activated and inhibited, were increased in T_ND recipients early post-HCT compared to TR. Like NK cells, most B cells except for plasmablasts and plasma cells were removed by CD45RA-depletion of PBSC. Immature, naïve, and memory B cells were reduced very early post-T_ND HCT, but substantial B cell recovery occurred within the first 2-3 months.

Conclusions: We present the first detailed description of lymphocyte reconstitution after T_ND HCT. Despite the major effect CD45RA-depletion has on graft composition-removing all T_N and T_{EMRA} from the T cell compartment and most NK and B cells-NK and B cell recovery occurred early, and by 1 year post-HCT there was little difference in lymphocyte populations between T_ND and TR. This is consistent with hematopoietic progenitors, not mature lymphocytes, as the source of B and NK cell reconstitution. Among T cells, the most striking findings were an increase in activated T_{CM} and T_{EM} CD4⁺ and CD8⁺ T cells early after T_ND HCT, and increased ratios of activated T cells to Treg. Although elevated blood T effector/Treg

ratios have been associated with cGVHD, cGVHD was uncommon and steroid-responsive after T_ND HCT. Whether increased activated T cells reflect a reduction in Treg-mediated suppression after HCT, bystander T cell activation, activation by antigen (e.g., alloantigen, latent herpes viruses, microbiota), or multiple factors are questions we are now investigating. We are also exploring associations between lymphocyte populations early post-T_ND HCT and clinical outcomes; these could serve as biomarkers to facilitate early intervention to prevent or treat GVHD, relapse, or infection.

Disclosures Shlomchik: BlueSphere Bio: Current Employment, Current holder of stock options in a privately-held company, Membership on an entity's Board of Directors or advisory committees, Patents & Royalties, Research Funding; Orca Bio: Consultancy, Current holder of stock options in a privately-held company. **Newell:** Immunoscape: Current equity holder in private company, Membership on an entity's Board of Directors or advisory committees; Neogene Therapeutics: Current equity holder in private company, Membership on an entity's Board of Directors or advisory committees; Nanostring Technologies: Membership on an entity's Board of Directors or advisory committees. **Bleakley:** Orca Bio: Consultancy; Miltenyi Biotec: Research Funding.



<https://doi.org/10.1182/blood-2023-187389>